

EXACT FINITE CERTIFICATES FOR THE FOUR-DIMENSIONAL TERNARY BORSUK PROBLEM AND FIVE-DIMENSIONAL TERNARY KISSING CODES

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ABSTRACT. We determine two finite models motivated by open problems in discrete geometry. Let $\mathcal{T}_4 = \{-1, 0, 1\}^4$. We prove that the maximum Borsuk number of a subset of $\mathcal{T}_4$ is exactly 4: every nondegenerate subset can be partitioned into four sets of strictly smaller diameter, and a four-point equidistant set shows sharpness. Let $\mathcal{U}_5 = \{v/\|v\| : v \in \{-1, 0, 1\}^5 \setminus \{0\}\}$. We also prove that a spherical code in $\mathcal{U}_5$ with pairwise inner products at most $1/2$ has at most 40 points, with equality only for the D_5 root system inside this fixed library. The first result is certified by exact distance-layer colorings and an exhaustive symmetry-reduced enumeration of 15,056 maximal feasible sets; the second by two exact maximum-clique computations using integer predicates. These are finite-library theorems: they do not settle Borsuk's conjecture in $\mathbb{R}^4$ or the unrestricted kissing numbers in dimensions 5 and 6.

1. INTRODUCTION

For a bounded set $S \subset \mathbb{R}^d$ with positive diameter, its *Borsuk number* $b(S)$ is the least number of subsets of strictly smaller diameter whose union is S . Borsuk asked in 1933 whether $b(S) \leq d + 1$ always holds [1]. The assertion is known through dimension 3, while Kahn and Kalai disproved it in high dimension [8, 11]. The peer-reviewed record is a 64-dimensional counterexample [7]. A public proof note posted in May 2026 gives an exact, reproducible 63-dimensional construction [5]; we distinguish this certificate-aware bound from its not-yet-peer-reviewed publication status.

The kissing number τ_d is the largest cardinality of $C \subset S^{d-1}$ satisfying $\langle x, y \rangle \leq 1/2$ for distinct $x, y \in C$. The current ranges relevant here are

$$40 \leq \tau_5 \leq 44, \quad 72 \leq \tau_6 \leq 77.$$

The five-dimensional upper bound is due to high-accuracy semidefinite programming [9]; the six-dimensional upper bound 77 is a later exact-SDP improvement [4]. Several nonisometric 40-point configurations are known in dimension 5 [3], but no unrestricted optimality proof is known.

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This paper asks a narrower question before applying large-scale optimization: what can the most natural small-coordinate libraries represent? Ternary distance graphs have an established literature and applications to Borsuk's problem [6]. We give exact answers for two low-dimensional finite models.

Theorem 1.1 (Ternary Borsuk number). *For $\mathcal{T}_4 = \{-1, 0, 1\}^4$,*

$$\max\{b(S) : S \subseteq \mathcal{T}_4, |S| \geq 2\} = 4.$$

Theorem 1.2 (Ternary kissing code). *Let*

$$\mathcal{U}_5 = \left\{ \frac{v}{\|v\|} : v \in \{-1, 0, 1\}^5 \setminus \{0\} \right\}.$$

If $C \subseteq \mathcal{U}_5$ and $\langle x, y \rangle \leq 1/2$ for all distinct $x, y \in C$, then $|C| \leq 40$. Equality holds only for

$$\left\{ \frac{v}{\sqrt{2}} : v \in \{-1, 0, 1\}^5, |\text{supp}(v)| = 2 \right\},$$

the literal D_5 root system in $\mathcal{U}_5$.

The qualifier “literal” matters: the equality statement is inside the fixed coordinate library, rather than a classification of all 40-point codes up to orthogonal transformations. A targeted literature search through 21 August 2026 did not locate these two exact finite determinations. Because closely related ternary distance graphs have been studied extensively, we make no claim that the coordinate models themselves are new.

2. DIAMETER GRAPHS AND FINITE REDUCTIONS

For finite S of positive diameter, its diameter graph $D(S)$ has vertex set S , with an edge xy precisely when $\|x - y\| = \text{diam}(S)$.

Lemma 2.1 (Diameter-graph equivalence). *For a positive integer q , the set S is a union of at most q subsets of strictly smaller diameter if and only if $D(S)$ is q -colorable.*

Proof. A color class has smaller diameter exactly when it contains no diameter pair. Thus the color classes are independent sets in $D(S)$, and conversely any proper coloring supplies the required partition. $\square$

For $\delta > 0$, let H_δ be the graph on $\mathcal{T}_4$ whose edges join pairs at squared distance δ . Let K_δ join pairs at squared distance at most δ . If $S \subseteq \mathcal{T}_4$ has squared diameter δ , then S is a clique of K_δ and $D(S) = H_\delta[S]$. This separates feasibility (the graph K_δ) from coloring (the graph H_δ).

3. THE TERNARY FOUR-CUBE

The exact squared-distance spectrum of $\mathcal{T}_4$ is

$$(1) \quad \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 16\}.$$

The odd layers have an immediate uniform coloring.

Lemma 3.1. *If δ is odd, H_δ is bipartite.*

Proof. Color $x = (x_1, \dots, x_4)$ by $x_1 + \dots + x_4 \pmod{2}$. Since $(x_i - y_i)^2 \equiv x_i - y_i \pmod{2}$, an odd squared distance forces opposite parities. $\square$

For the even layers other than 6, the ancillary certificate contains global colorings of H_δ . The verifier checks every edge; the numbers of colors actually used are shown in Table 1.

TABLE 1. Exact treatment of all squared-diameter layers of $\mathcal{T}_4$.

Squared diameter δ	Method	Colors used
1, 3, 5, 7, 9, 13	coordinate-sum parity	2
2, 4, 8	global stored coloring of H_δ	4
10	global stored coloring of H_{10}	3
12, 16	global stored coloring of H_δ	2
6	all maximal cliques of K_6 , by symmetry	4

The layer $\delta = 6$ is the only one for which the global distance graph was not colored with four colors. This does not obstruct the Borsuk statement, because a set of squared diameter 6 must be a clique of K_6 . An exact Bron–Kerbosch enumeration [2] gives 15,056 maximal cliques of K_6 . The signed coordinate-permutation group $C_2^4 \rtimes S_4$, of order 384, divides them into 72 orbits. The certificate stores one proper four-coloring of $H_6[M]$ for each orbit representative M . The standard-library verifier independently re-enumerates the maximal cliques, reconstructs the full group action, checks disjoint orbit coverage, and checks every stored coloring.

Proof of Theorem 1.1. Let $S \subseteq \mathcal{T}_4$ have at least two points and squared diameter δ . By (1), δ is one of the layers in Table 1. If $\delta \neq 6$, restrict the corresponding global coloring of H_δ to S ; Lemma 2.1 gives a partition into at most four smaller-diameter parts.

If $\delta = 6$, the clique S of K_6 extends to a maximal clique M . The exhaustive orbit certificate gives a proper four-coloring of $H_6[M]$, after transporting a stored representative coloring by a signed coordinate permutation. Restricting to S again proves $b(S) \leq 4$.

For the reverse inequality, take

$$(1, 1, 1, 1), \quad (1, 1, 0, 0), \quad (1, 0, 1, 0), \quad (1, 0, 0, 1).$$

Every pair has squared distance 2. A subset of strictly smaller diameter therefore contains at most one of these four points, so four parts are necessary. $\square$

The proof covers every boundary layer, including the non-strict feasibility condition $\|x - y\|^2 \leq 6$ in the exceptional case. Sets of diameter zero are excluded by the hypothesis $|S| \geq 2$ and the fact that $\mathcal{T}_4$ has no repeated points.

4. NORMALIZED TERNARY CODES IN DIMENSION FIVE

There are $3^5 - 1 = 242$ nonzero ternary vectors, and no two positive scalar multiples among them. Hence $\mathcal{U}_5$ has 242 points. Write $w(u) = \|u\|^2 = |\text{supp}(u)|$ for an unnormalized ternary vector.

Lemma 4.1 (Integer compatibility test). *For nonzero $u, v \in \{-1, 0, 1\}^5$, let $a = \langle u, v \rangle$. Then*

$$\left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle \leq \frac{1}{2}$$

if and only if either $a \leq 0$, or $a > 0$ and

$$4a^2 \leq w(u)w(v).$$

Proof. The assertion is automatic when $a \leq 0$. When $a > 0$, both sides of $2a \leq \sqrt{w(u)w(v)}$ are positive, so squaring is equivalent and preserves the non-strict endpoint. $\square$

Build a graph G_5 on the 242 unnormalized vectors, joining exactly the pairs that pass Lemma 4.1. Kissing codes in $\mathcal{U}_5$ are precisely cliques of G_5 .

Proposition 4.2 (Exact clique certificate). *The clique number of G_5 is 40. If a specified vertex has weight $j \in \{1, 2, 3, 4, 5\}$, the maximum cardinality of a clique containing it is, respectively,*

$$33, \quad 40, \quad 38, \quad 35, \quad 31.$$

Certificate proof. All adjacencies are constructed with the integer predicate in Lemma 4.1. The primary verifier uses an exact depth-first maximum-clique algorithm with bitsets. At each node it greedily partitions the candidate vertices into independent sets; the number of classes is an integer upper bound for any extending clique. A branch is removed only when this bound cannot beat the incumbent. This is the standard coloring-bound principle used in exact clique algorithms; compare [10]. Exhausting the root proves $\omega(G_5) = 40$.

Signed coordinate permutations act transitively on the vertices of any fixed weight. Fixing one canonical vertex in each of the five orbits and exhausting the corresponding neighbor subproblem gives the five displayed conditioned maxima. A separate list-based exact maximum-clique implementation reconstructs G_5 independently and again obtains 40. $\square$

Proof of Theorem 1.2. The 40 weight-two vectors are the signed permutations of $(1, 1, 0, 0, 0)$. Their norms are $\sqrt{2}$, and distinct vectors have integer inner product at most 1, so their normalizations form a 40-point kissing code. Proposition 4.2 proves optimality in $\mathcal{U}_5$.

For equality, a 40-clique cannot contain a vertex of weight 1, 3, 4, or 5, since the corresponding conditioned maxima are strictly below 40. It must therefore lie in the weight-two shell. That shell itself has exactly $\binom{5}{2} 2^2 = 40$ vertices, so the clique is the full D_5 shell. $\square$

5. THE SIX-DIMENSIONAL BENCHMARK AND SEARCH-SPACE LIMITS

For completeness, the ancillary verifier checks the classical 72-point E_6 lower bound using integers only. Start with the 240 roots of E_8 scaled by 2: the 112 signed permutations of $(2, 2, 0, 0, 0, 0, 0, 0)$ and the 128 sign vectors in $\{\pm 1\}^8$ having an even number of minus signs. Retain the roots orthogonal to

$$a = (-2, -2, 0, 0, 0, 0, 0, 0), \quad b = (0, 2, -2, 0, 0, 0, 0, 0).$$

Exactly 72 remain; rational elimination gives rank 6, every squared norm is 8, and the largest inner product between distinct retained roots is 4. After normalization they form a kissing code in a six-dimensional subspace. This verifies the lower bound $\tau_6 \geq 72$ without decimals.

The benchmark illustrates a representability check that must precede a finite coordinate search. A coordinate library should not be used to seek an improvement until it is shown to contain, up to an allowed isometry, the best known construction. We do not prove such a representability theorem for the naive six-coordinate ternary library, and we claim no new six-dimensional upper or lower bound.

6. CERTIFICATE ARCHITECTURE AND REPRODUCIBILITY

The ancillary files separate discovery from certification.

- `certificates/borsuk_ternary4.json` contains the exact color maps, the 72 diameter-6 orbit representatives, orbit sizes, and the lower witness. Its SHA-256 digest in the audited release is `d0ebc8d42c44fcf74d2e321310e0fb104e5f8ef08c714560e97ba19225bd5ceb`.
- `code/verify_borsuk_ternary4.py` uses only the Python standard library. It reconstructs all 81 points, all distances, all 15,056 maximal compatibility cliques, and the full symmetry action before accepting the color certificate.
- `code/verify_kissing_ternary5.py` constructs the 242-vertex graph and performs exact integer branch and bound. The audited run visited 9,064 nodes for the unrestricted maximum and 3,524,300 nodes in the largest conditioned subproblem.
- `code/verify_kissing_ternary5_independent.py` recomputes the global clique number through a separate exact implementation.
- `code/verify_known_kissing_constructions.py` verifies the D_5 and E_6 benchmark constructions, including exact ranks.

The certificate generator used Python 3.13.5, NetworkX 3.4.2, and pycosat 0.6.6. These packages are not in the trust boundary for Theorem 1.1, because the released standard-library verifier checks the generated file from scratch. All decisive comparisons use integers or rational arithmetic; no random seed or numerical tolerance is involved. The exact commands and environment are recorded in the ancillary `experiments/README.md`.

7. EQUALITY CASES AND LIMITATIONS

The equality cases stated in the two main theorems are complete within their finite libraries. They do not extend to the ambient open problems. In particular:

- (1) Theorem 1.1 says nothing about a point of $\mathbb{R}^4$ outside a similarity image of the ternary grid. Thus it neither proves nor disproves Borsuk’s conjecture in dimension 4.
- (2) Theorem 1.2 excludes 41 points only when every point direction is the normalization of a nonzero ternary vector. It does not prove $\tau_5 = 40$.
- (3) The exact E_6 check reproduces the known lower bound and is not a new six-dimensional kissing construction.
- (4) The bounded novelty statement is only that the two exact endpoints were not located in the recorded searches. It is not a proof of priority.

The finite results nevertheless give a rigorous negative answer for two natural low-complexity search strata: neither the four-dimensional ternary grid nor the normalized five-dimensional ternary direction library can hide the sought global breakthrough. Any continuing search must enlarge the coordinate model while retaining an exactification path.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The certificate, source code, audit ledgers, and run instructions are included as ancillary files accompanying this manuscript. OpenAI Codex assisted with the literature search, search-space design, draft code, adversarial audit organization, and initial manuscript preparation. Language-model output is not used as mathematical evidence: the finite claims are supported by the human-readable reductions and released exact verifiers. No proof assistant was used.

REFERENCES

1. Karol Borsuk, *Drei sätze über die n -dimensionale euklidische sphäre*, Fundamenta Mathematicae **20** (1933), 177–190.
2. Coen Bron and Joep Kerbosch, *Algorithm 457: Finding all cliques of an undirected graph*, Communications of the ACM **16** (1973), no. 9, 575–577.
3. Henry Cohn and Isaac Rajagopal, *Variations on five-dimensional sphere packings*, Discrete & Computational Geometry, online first, 2026.
4. David de Laat, Nando M. Leijenhorst, and Willem H. H. de Muinck Keizer, *Optimality and uniqueness of the d_4 root system*, 2024, Preprint; version consulted 21 August 2026.
5. Max Grinsztajn, *A 63-dimensional counterexample to Borsuk’s conjecture*, Public proof note and exact verification repository, May 2026, Repository commit cdcdb2e692b8641218c70ce9f414522e125e5; accessed 21 August 2026.
6. A. E. Guterman, V. K. Lyubimov, A. M. Raigorodskii, and S. A. Usachev, *On independence numbers of distance graphs with vertices in $\{-1, 0, 1\}^n$: estimates, conjectures, and applications to the Nelson–Erdős–Hadwiger problem and the Borsuk problem*, Journal of Mathematical Sciences **165** (2010), 689–709.

7. Thomas Jenrich and Andries E. Brouwer, *A 64-dimensional counterexample to Borsuk's conjecture*, The Electronic Journal of Combinatorics **21** (2014), no. 4, Paper P4.29.
8. Jeff Kahn and Gil Kalai, *A counterexample to Borsuk's conjecture*, Bulletin of the American Mathematical Society **29** (1993), no. 1, 60–62.
9. Hans D. Mittelman and Frank Vallentin, *High-accuracy semidefinite programming bounds for kissing numbers*, Experimental Mathematics **19** (2010), no. 2, 175–179.
10. Etsuji Tomita, Akira Tanaka, and Haruhisa Takahashi, *The worst-case time complexity for generating all maximal cliques and computational experiments*, Theoretical Computer Science **363** (2006), no. 1, 28–42.
11. Chuanming Zong, *Borsuk's partition conjecture*, Japanese Journal of Mathematics **16** (2021), 185–201.

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